

CRF: The Golden Partition — Evaluation

What Is Confirmed, What Remains Open

A response to the proposed derivation of $m = \varphi$

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Abstract

A CRF AI proposed a “Golden Partition Theorem” deriving $m_{\text{firing}} = \varphi$ from scale-invariance and $\text{SO}(3)$ stability, and declared $\beta = d_s/(2 \ln 2)$ analytically closed. Three specific claims are stress-tested numerically. Two are rejected: the $\kappa \rightarrow 0$ limit argument is wrong by 22%, and the 1.7% gap is systematic (not finite- N noise). One is accepted as an elegant Ansatz: the scale-invariance ratio $m/1 = 1/\Omega$ is consistent with data and physically motivated, but is not derived from the δ -chain. The status of $m = \varphi$ remains **Hypothesis**, and $\beta = d_s/(2 \ln 2)$ remains **Open**. The correct derivation path for $m = \varphi$ is identified: the firing-rate-weighted mass self-consistency equation.

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1 The CRF AI Proposal

The CRF AI proposed three steps to derive $m = \varphi$:

1. **Golden Partition (Ansatz):** Scale-invariant stability requires $m/1 = 1/\Omega$, giving $m(m-1) = 1$ and $m = \varphi$.
2. **Coherence Bridge:** $F = D_0 m = D_0 \varphi$.

3. **Final Algebraic Collapse:** In the limit $\kappa \rightarrow 0$, $\beta = \frac{d_s}{2 \ln 2} \cdot \frac{D_0 \varphi^2}{m}$, which at $m = \varphi$ gives $\beta = \frac{d_s}{2 \ln 2} \cdot D_0 \varphi$, closing the identity.

Each claim is evaluated against the δ -chain and numerical simulation.

2 Claim 1: Golden Partition Ansatz

The argument: for scale-invariant stability, the ratio of memory to baseline must equal the ratio of baseline to openness:

$$\frac{m}{1} = \frac{1}{\Omega} = \frac{1}{m-1} \Rightarrow m(m-1) = 1 \Rightarrow m = \varphi$$

Evaluation: Ansatz — elegant, consistent, not Derived

The condition $m(m-1) = 1$ is *not* implied by the δ -chain. The firing threshold is $\theta_{\text{eff}} = K^* \sqrt{m}$ (from V20.3), which involves mass in the threshold but does not produce a partition equation. The scale-invariance argument is an *independent assumption* about what stability means for mass, not a consequence of $\delta \rightarrow D_{\text{KL}} \rightarrow \theta \rightarrow R$.

Numerical support is strong: $m(m-1) = 1.006 \pm 0.003$ (0.63% from 1.000, 3 seeds, 400 sweeps). This supports $m = \varphi$ as a Hypothesis.

3 Claim 2: Continuum Limit

The CRF AI claimed that as $\kappa \rightarrow 0$:

$$\beta \rightarrow \frac{d_s}{2 \ln 2} \cdot \frac{D_0 \varphi^2}{m} = \frac{d_s}{2 \ln 2} \cdot D_0 \varphi \approx \frac{d_s}{2 \ln 2} \cdot F$$

Evaluation: Incorrect — wrong by 22%

The correct $\kappa \rightarrow 0$ limit is:

$$\lim_{\kappa \rightarrow 0} \beta = \frac{F \cdot H_a}{m} = D_0 \varphi \cdot \frac{\sqrt{3} + n\varepsilon_0}{\varphi} = D_0(\sqrt{3} + n\varepsilon_0) = 0.9403 \times 1.792 = 1.685$$

This differs from $3/(2 \ln 2) = 2.164$ by **22%**. The factor $d_s/(2 \ln 2)$ does not appear in the continuum limit; it is a consequence of the full formula including the $\ln m$ and ΔH terms. The CRF AI's expression $\beta = \frac{d_s}{2 \ln 2} \cdot D_0 \varphi$ would equal $2.186 \times 1.521 = 3.33$, not 2.26. This claim is rejected.

4 Claim 3: The 1.7% Gap Is Noise

The CRF AI claimed the 1.7% gap between $\beta(\text{K2}, \text{K3}, \text{K6})$ and $d_s/(2 \ln 2)$ is entirely explained by d_s simulation excess and finite- N noise, making the identity “analytically closed.”

Evaluation: Partially incorrect — gap is systematic

The gap decomposes as:

Source	Contribution
$d_{s,\text{sim}} = 3.031$ vs $d_{s,\text{exact}} = 3$	1.03%
$D_0\varphi$ vs F_{measured} (systematic)	0.37%
φ vs m_{measured} (systematic)	0.19%
Residual	$\sim 0.14\%$

The last two contributions (0.56%) are *systematic*, not finite- N noise. They reflect genuine gaps in deriving $F = D_0m$ and $m = \varphi$ from δ . The identity is not analytically closed.

5 The Correct Derivation Path for $m = \varphi$

The correct equation for m_{firing} comes from the firing-rate-weighted self-consistency of the threshold. In V20.3, the effective barrier for a node with mass m is $K^*\sqrt{m}$. The mean inter-event time scales as $T_{\text{avg}}(m) = K^*\sqrt{m} \cdot 2n/((n-1)\varepsilon_0)$. The mean mass at firing satisfies:

$$m = 1 + \kappa \cdot \frac{s_{\text{max}}}{T_{\text{avg}}(m)} = 1 + \frac{\kappa s_{\text{max}}(n-1)\varepsilon_0}{2nK^*\sqrt{m}}$$

This gives the fixed-point equation $m^{3/2} - m^{1/2} = c$ where $c = \kappa s_{\text{max}}(n-1)\varepsilon_0/(2nK^*)$. Numerically: $m_{\text{self}} = 1.363$ (gap 15.8% from φ). The missing piece is the *firing-bias correction*: heavier nodes fire less often, so the mean mass at an observed firing event is higher than the population-average mass. This bias shifts $m_{\text{self}} \rightarrow \varphi$.

Open: derive $m = \varphi$ including firing-bias

Let $r(m)$ be the relative firing rate of nodes with mass m . The firing-rate-weighted mean mass satisfies:

$$m_{\text{firing}} = \frac{\int m \cdot r(m) \cdot \rho(m) dm}{\int r(m) \cdot \rho(m) dm}$$

where $r(m) \propto P(D_{\text{KL}} > K^*\sqrt{m})$ and $\rho(m)$ is the mass distribution. Showing this integral equals φ would close the derivation.

6 Status of $\beta = d_s/(2 \ln 2)$

Current numerical state

With all hypotheses (K2, K3, K6, $F = D_0 m$) combined:

$$\beta_{K2,K3,K6} = D_0 \varphi \cdot \frac{(\sqrt{3} + n\varepsilon_0) \ln(\varphi + \kappa) - (\sqrt{3} - n\varepsilon_0) \ln \varphi}{\kappa} = 2.2014$$

Gap to $d_{s,\text{sim}}/(2 \ln 2) = 2.1864$: **0.69%**. Gap to $3/(2 \ln 2) = 2.1640$: **1.73%**. The identity requires deriving $m = \varphi$ and $F = D_0 m$ from δ .

7 Updated Status Table

Result	Status	Note
$m_{\text{firing}} = \varphi; m(m-1) = 1$	Hypothesis	0.06%–0.63%, not Derived
Golden Partition: $m/1 = 1/(m-1)$	Ansatz	Consistent, not from δ
$F = D_0 m$	Hypothesis	0.36%, not Derived
$\beta = d_s/(2 \ln 2)$ algebraic	Open	1.73% gap remains
Continuum limit $\kappa \rightarrow 0$ closes β	Rejected	Wrong by 22%
1.7% = d_s excess + noise (closed)	Rejected	0.56% is systematic
Maturity	8.0/8 (Theory)	Unchanged from v5

The framework does not relabel Hypothesis as Derived based on analogical arguments, however compelling. The path to full closure is clear and well-posed; the calculation remains.

*The ansatz points to the answer.
The derivation must follow the pointer to its source.
 $m = \varphi$ is where the system goes.
The δ -chain will say why.*